



GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
Department of Computer Science and Engineering

Major Project Review-1

**FixMyCity: Smart Civic Issue Reporting and Urban
Decision Support System**

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ABSTRACT

- Rapid urbanization has increased civic issues demands intelligent civic management beyond traditional complaint registration and tracking systems
- Existing systems focus on complaint registration and tracking, lacking intelligent analysis and data driven support for municipal decision making
- FixMyCity enables civic reporting and explores the use of GIS and ML to identify hotspots, recurring infrastructure failures, and citywide trends
- The platform supports environmental analytics predict pollution, seasonal issues, groundwater availability, and recommend sustainable municipal actions.
- The proposed system aims to improve municipal efficiency, transparency, citizen participation, and environmental sustainability by enabling proactive governance

LITERATURE SURVEY

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Shortcomings/ Gaps in The Paper
[1]	CitySolution: A Complaining Task Distributive Mobile Application for Smart City Corporation using Deep Learning (2024)	Farhatun Shama, Abdul Aziz, Lamisa Bintee Mizan Deya	The paper proposes a smart city complaint management application that enables citizens to report civic issues through a mobile platform. Complaint images are processed using deep learning techniques to automatically classify the reported issue.	does not support intelligent duplicate complaint detection, GIS hotspot analysis, or predictive analytics for recurring civic problems.
[2]	Smartreporter – A Crowdsourced Complaint Resolution System (2026)	Kurian Binu et al.	The system employs YOLO-based object detection to recognise infrastructure defects from uploaded images and provides real-time complaint tracking	The system mainly concentrates on complaint reporting and monitoring without citizen engagement, identifying spatial hotspots, perform duplicate complaint merging.

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Gaps / Limitations
[3]	Global Road Damage Detection Challenge: State-of-the-Art Solutions	Deeksha Arya, Hiroya Maeda, Sanjay Kumar Ghosh, Durga Toshniwal, Hiroshi Omata, Takehiro Kashiya, Yoshihide Sekimoto	The study evaluates multiple deep learning object detection models for automatic road damage identification using large-scale benchmark datasets.	The research focuses exclusively on road damage detection and does not consider other urban infrastructure
[4]	Sentence-BERT: Sentence Embeddings using Siamese BERT Networks	Nils Reimers, Iryna Gurevych	OCR-based ingredient extraction from food labels and barcodes. Hybrid AI engine combining Retrieval-Augmented Generation (RAG) with a fine-tuned LLaMA-2 model for ingredient safety analysis	Only focuses on analyzing packaged food labels and providing personalized dietary guidance. It does not support reporting citizen grievances.

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Gaps / Limitations
[5]	Learning Transferable Visual Models From Natural Language Supervision (CLIP)	Alec Radford , Jong Wook Kim , Chris Hallacy , Aditya Ramesh , Gabriel Goh , Sandhini Agarwal , Girish Sastry , Amanda Askell	Contrastive learning between image-text pairs using dual encoders (image + text). Evaluated via zero-shot transfer on tasks like ImageNet	Requires domain-specific prompts or fine tuning for regulatory interpretation
[6]	Software Platforms for Smart Cities: Concepts, Requirements, Challenges and a Unified Reference Architecture	Eduardo F. Nakamura, Artur Ziviani, Eduardo Santana et al.	The paper presents a comprehensive reference architecture for smart city platforms by integrating Internet of Things (IoT), cloud computing, big data analytics, Geographic Information Systems (GIS), and urban services into a unified framework.	Although the paper provides a strong architectural foundation for smart city platforms, it remains largely conceptual and does not propose decision support mechanisms.

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Gaps / Limitations
[7]	GeoAI: Where Machine Learning and Big Data Converge in GIScience	Wenwen Li	This paper introduces GeoAI as an emerging research field that combines artificial intelligence, machine learning, and geospatial analytics for processing large-scale spatial datasets.	The paper primarily focuses on the theoretical foundations and applications of GeoAI rather than developing a complete smart governance solution
[8]	AI-Based Smart Civic Complaint Management System	Suganthi C. et al.	The proposed system applies artificial intelligence techniques to classify and manage civic complaints submitted by citizens through digital platforms.	Paper concentrates on complaint processing after incidents occur and lacks predictive capabilities for seasonal civic problems

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Gaps / Limitations
[9]	DeepAir: A Hybrid Deep Learning Framework for Air Quality Forecasting	Shengdong Du; Tianrui Li; Yan Yang; Shi-Jinn Horng	A hybrid deep learning framework that combines meteorological observations, historical pollution measurements, and spatial information to forecast air quality with multiple deep learning architectures	Does not consider citizen-generated complaints, municipal infrastructure information, or explainable policy recommendations
[10]	Machine Learning Models for Groundwater Level Prediction and Uncertainty Analysis	Dawit Kanito; Mohammed Benaafi; Husam Musa Baalousha	review summarizes recent machine learning approaches for groundwater level prediction using Random Forest, Support Vector Regression, Artificial Neural Networks, LSTM, and XGBoost models.	The reviewed studies mainly focus on groundwater prediction as an isolated hydrological problem and do not integrate urban infrastructure information.

Ref. No.	Title of the Paper & Year	Authors	Methodology / Approach	Observed Gaps / Limitations
[11]	A Unified Approach to Interpreting Model Predictions (SHAP)	Scott M. Lundberg, Su-In Lee	A unified intelligence framework that interprets machine learning predictions by assigning contribution scores to individual input features.	SHAP provides explainability for machine learning models but does not offer domain specific decision-support mechanisms or urban governance applications.

Proposed System

- Build a MERN-based civic management platform with WhatsApp chatbot integration for issue reporting.
- Utilize Computer Vision, OCR, NLP, and Speech-to-Text to process multimedia complaints.
- Implement Machine Learning for issue prioritization, hotspot detection, seasonal prediction, and groundwater estimation.
- Integrate GIS and public datasets to visualize complaints, environmental conditions, and city-wide risk zones.
- Develop an AI-powered Decision Support System providing explainable recommendations and predictive analytics.

- Enable transparent complaint tracking, citizen participation, and intelligent resource allocation for municipal authorities.
- Estimate groundwater recharge potential using rainfall, land use, vegetation, and groundwater datasets.
- Detect air pollution hotspots and generate AI-powered mitigation recommendations for municipal authorities.

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Thank You!!